The year is 2080, and our oceans are louder than they have been in centuries.

Not with ships or war - but with signals.

At the headquarters of the Oceanic Research Coalition (ORC), banks of monitors pulse with a pattern no one can explain. Deep beneath the Pacific gyres, below shipping lanes and fiber-optic cables, below even the migratory routes of whales, a rhythmic sonar beacon is repeating every seventy-three seconds.

It is not natural. Something is waking up.

The signal originates from a trench system long dismissed as geological noise - a zone sailors once called *The Abyssal Veil*. Satellite bathymetry shows a labyrinth of chasms spiraling down nearly eleven kilometers, their walls threaded with ancient coral growth that should not survive at such depths. Yet faint bioluminescent trails shimmer there like submerged constellations.

For centuries, Pacific island lore spoke of the Siren Engineers, aquatic beings neither myth nor machine, said to shape ocean currents the way shepherds guide flocks. They were guardians of thermohaline circulation, sculptors of reef nurseries, architects of migratory highways.

According to legend, when climate instability first threatened the seas, the Sirens built vast current-modulating engines within the deepest trenches. These devices did not burn fuel or split atoms. They sang - resonant harmonic pulses that tuned salinity gradients and temperature flows, stabilizing planetary systems from below.

Then, as abruptly as they appeared in folklore, the Sirens vanished.

The oceans balanced themselves for a time.

Until now.

In the past six months:

- Atlantic overturning currents have begun to oscillate unpredictably.
- Coral spawning cycles are desynchronizing across hemispheres.
- Dead zones are forming where oxygen levels once thrived.
- Whale pods have altered migration paths as if avoiding unseen turbulence.

Every anomaly triangulates back to The Abyssal Veil.

The sonar pulses have intensified. Not random - structured. Layered harmonics embedded within the signal resemble modulation patterns. Engineers at ORC confirm it: the pulses are activating something.

Your team has the most advanced deep-sea explorer ever constructed. Your objective is clear:

1. Descend into The Abyssal Veil and recover the artifacts hidden within its depths.

2. Restore power to the Siren network and decode the message hidden within its signal.

3. Use the decoded message to identify and make the correct offering to the Siren.

Failure will not mean an explosion.

It will mean imbalance - slow, planetary unraveling.

As you breach the Veil's perimeter, ordinary sonar dissolves into interference. The trench walls are not barren rock but intricate lattices - calcified coral structures interwoven with metallic filaments, half-organic, half-engineered.

Bioluminescent organisms pulse in sync with the sonar rhythm.

The deeper you descend, the clearer it becomes: this is not ruin.

It is infrastructure.

Towering pylons rise from the trench floor, spiraled like nautilus shells, each embedded with crystalline membranes that vibrate faintly. Between them stretch conduits resembling reef growth but humming with kinetic energy.

The Siren Engineers did not merely build machines.

They cultivated them.

At 10,842 meters, your sensors lock onto the epicenter.

A colossal structure rests within a cavern dome - a harmonic engine the size of a city block. Rings of unknown alloy rotate slowly around a luminous core. Every seventy-three seconds, the core emits the pulse detected on the surface.

But the harmonics are fractured.

Instead of a stabilizing chord, the engine produces dissonance. Competing frequencies ripple outward, colliding in the water column above. The destabilization of global currents is not sabotage, it is misalignment.

Something in the system is out of tune.

You are able to isolate three sub-nodes feeding into the central engine:

- A thermal regulator compensating for ocean warming.
- A salinity modulator correcting freshwater imbalance.
- A biospheric synchronizer linked to marine reproductive cycles.

One, or more, has drifted from calibration.

To restore balance, you must:

- Navigate the narrow trench and restore power to the Siren network.
- Decode harmonic signatures left by the Siren architecture using Morse code.
- Recover artifacts to aid ongoing research and uncover what happened in The Abyssal Veil.

If you fail, the engine may amplify instability instead of calming it.

If you succeed, the oceans will breathe evenly once more.

As you approach the control nexus, a final anomaly appears.

The pulse pattern shifts.

It adapts.

The system is not merely malfunctioning.

It is responding.

To you.

The Siren Engineers may be gone.

Or they may be listening.

And in the silent pressure of the deep, the future of Earth's oceans now rests on whether you can learn to sing back.

Mission Objectives

1. Message in a Bottle

Objective: Recover three bottles to discover the locations of hidden artifacts inside the cave.

1. Retrieve three (3) bottles and bring them to the surface.
2. Read the messages inside the bottles to find the X and Y coordinates of the three artifacts.
3. Use the coordinates to locate and retrieve the artifacts from the corresponding cubby inside the cave.
4. Place the retrieved artifacts on the display shelf near the mission entrance.

Important Notes:

- Each artifact will be no larger than 6 in. × 6 in. × 6 in.
- Teams must recover and decode the messages before attempting to retrieve the artifacts.

2. Shipwreck Cleanup

Objective: Clean up the kelp forest and reunite a lost seahorse with its shoal.

1. Remove the trash from the coral, located within the kelp forest.
2. Transport the trash to the designated garbage collection zone and deposit it into the collection chute.
3. Retrieve the lost seahorse from the cave.
4. Transport the seahorse to the kelp forest and place it with the shoal.

3. Trapped Treasure

Objective: Open the treasure chest and recover the crown.

1. Locate the trident blocking the treasure chest inside the cave.
2. Rotate the trident 90 degrees to trigger the chest lid to open and raise the crown on its platform.
3. Retrieve the crown and bring it to the surface.

Important Notes:

- The trident must be pushed far enough to engage a latch. If it is not pushed far enough, it will spring back to its original vertical position and the chest will not open.
- Teams may keep the crown at the end of the event if they successfully retrieve it.

4. Restore Power

Objective: Restore power by routing an adapter through the trench and connecting it to the power terminal.

1. Using the provided prop, navigate the trench and route the cable through four (4) waypoints.
2. Confirm that each waypoint illuminates, indicating that the cable has been correctly routed.
3. Insert the power adapter into the power terminal along the back wall of the trench.

Mission Result: Successfully connecting the power adapter will be indicated by a light and will open the side entrance to the cave, providing an additional route into the cave.

5. Decode the Sound

Objective: Decode the siren signal activated when the cave's side entrance opens.

1. Connect your team's hydrophone to an audio system to listen to the siren signal.
2. Use hydrophone audio and Morse code to decode the message.
3. Record the decoded message for use in the *Make an Offering* mission.

Important Notes:

- NURC will provide a speaker at the mission table with a 3.5 mm audio input jack for connecting a hydrophone.
- Teams requiring a different audio connection or configuration must provide their own compatible speaker.

6. Make an Offering

Objective: Use the decoded siren message to identify and deliver the correct coin to the offering shell.

1. Use the decoded message from the *Decode the Sound* mission to identify the correct coin.
2. Retrieve the identified coin from the surface.
3. Transport the coin to the throne stage and place it inside the offering shell.

7. Snapper's Photo Op

Objective: Locate and document three hidden snapper figures using the ROV's onboard camera.

1. Search the pool for the three (3) snapper figures.
2. Position your ROV close enough for each figure to be clearly visible on the onboard camera's live video feed.
3. Notify the poolside judge when each figure has been located and is visible on the live feed.
4. Ensure each sighting is recorded on the deck scoresheet.

8. Identify the Starfish

Objective: Identify the starfish present in the designated area.

1. Observe the designated area using your ROV.
2. Identify the starfish you can see, using each starfish's color to distinguish it from the others.
3. Report the colors of the starfish you observe to the judge.

Important Notes:

- Starfish will change between zones between matches. Teams should report the starfish observed during their own match.

Mission Locations

Task	Task Location
Message in a bottle	Retrieve one bottle from each location: in cave, in kelp forest, in open water (deep end)
Shipwreck cleanup	Sloped end - kelp forest
Collect trapped treasure	Deep end - in cave
Restore power	Shallow end - in trench
Snapper's photo op	All zones - 3 total
Identify the starfish	All zones - 5 total
Decode the sound	All zones
Make an offering	Deep end - in cave

Point Breakdown

Mission	Task	Points
Message in a Bottle	Retrieve Bottle #1	10
	Retrieve Bottle #2	10
	Retrieve Bottle #3	10
	Retrieve correct Artifact #1	20
	Retrieve correct Artifact #2	20
	Retrieve correct Artifact #3	20
	Place correct artifact #1 on display shelf	15
	Place correct artifact #2 on display shelf	15
	Place correct artifact #3 on display shelf	15
Subtotal		135

Shipwreck Cleanup	Deposit all debris in garbage collection zone	20
	Locate lost seahorse	10
	Return seahorse to shoal	20
Subtotal		50
Collect the Trapped Treasure	Rotate trident to unlock chest	35
	Return crown to surface	25
Subtotal		60
Restore Power	Enter trench	5
	Thread cable through each waypoint (10 pts each, 4 waypoints). Points are totaled at mission end.	Max 40
	Successfully connect the power terminal.	35
Subtotal		80
Snapper's Photo Op	Locate Snapper #1	5
	Locate Snapper #2	5
	Locate Snapper #3	5
Subtotal		15
Count the Starfish	Starfish correctly identified (5 total, 10 points per)	Max 50
Decode the Sound	Correctly decode message	50
Make an Offering	Place any coin in shell	5
	Place correct coin in shell at throne	20

Total Maximum Score: 480

Game Specific Penalties

- Incorrect artifact retrieved: −10 per incorrect artifact on shelf

In addition to scorable points and game specific penalties, there are some general actions that will deduct points from a team's mission score. Please consult the Competition Handbook for more details.

Penalty	Penalty Points
Diver Assistance Penalty (ROV stuck but able to complete mission)	-20
Tether Penalty (person pulling tether to retrieve ROV)	-20 per infraction
Mission Prop Penalty (damaging field elements)	-5 per infraction
Diver Rescue (ROV unable to complete mission)	Mission End
Exceeding mission time limit (20 minute mission time does not include setup and tear down time)	-20/minute

Total points in the competition:

- Journal Paper 100 points
- Presentation 100 points
- Poster 25 points
- Video 25 points
- Mission 480 points

  Total points 730 points